

The $d = 2$ theorem for the chart amplitude $\eta e^{\beta s \xi}$

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Abstract

The amplitude-level hypotheses of unit 88 (axis Lipschitz bounds and the mixed second difference) are derived for $\Phi(u, v, s) = \eta(u, v) e^{\beta s \xi(u, v)}$ from Lipschitz and mixed-difference bounds on ξ, η themselves, which hold for C^2 functions. The rectangle difference of a product, $D[f g] = D[f] g_3 + (f_1 - f_0)(g_3 - g_1) + (f_2 - f_0)(g_3 - g_2) + f_0 D[g]$, and the identity $e^{t_3} - e^{t_1} - e^{t_2} + 1 = (e^{t_1} - 1)(e^{t_2} - 1) + (e^{t_3} - e^{t_1+t_2})$ give $|D[\eta e^{\beta s \xi}]| \leq C_2 u v (1 + s)^2 e^{5 \beta L s}$. Consequently the two-dimensional chart integral with equal exponents satisfies $Z(N) = N^{-p}(A \log N + B) + O(N^{-(p+\delta)}(1 + \log N))$ with $A = (k_1 k_2)^{-1} \eta(0, 0) S_p(\xi(0, 0))$; the n -form follows with $\frac{A}{2} \log n$. Zero sorry/axiom.

1 The things formalised

```
theorem amp_sub_le ( L M          s : ) (h : 0 < ) (hs : 0   s) (h : | | L)
  (h : | | L) (h : | | M) :
  | * Real.exp ( * s * ) - * Real.exp ( * s * ) |
  (| - | + M * ( * s) * | - |) * Real.exp ( * s * L)
theorem amp_mixed_le ... :
  | e * Real.exp ( * s * a) - e * Real.exp ( * s * a) - e * Real.exp ( * s * a)
  + e * Real.exp ( * s * a) |
  (K + 2 * K ^ 2 * + M * ^ 2 * K ^ 2 + M * * K) * (u * v) * ((1 + s) ^ 2
  * Real.exp (5 * * L * s))
noncomputable def chartAmp ( : ) ( : → → ) (u v s : ) : :=
  u v * Real.exp ( * s * u v)
theorem twoDChart_second_order ... (hu hv hu hv : Lipschitz in each variable)
  (hD hD : mixed second differences K * (u * v)) :
  (fun N : => twoDAmp b N h h k k (chartAmp )
    - N ^ (-p) * (twoDA p k k (chartAmp ) * Real.log N
    + twoDB b p k k (chartAmp ))) = 0[atTop]
  fun N : => N ^ (-(p + min (k : ) ^ 1 (k : ) ^ 1)) * (1 + Real.log N)
theorem twoDChart_A ( p : ) (k k : ) ( : → → ) :
  twoDA p k k (chartAmp )
  = 1 / ((k : ) * k) * ( 0 0 * weightedMass ( 0 0) (p - 1))
```

2 Roadblocks and resolutions

- `linarith` needs the product $\beta s L \geq 0$ as a named hypothesis to compare $2\beta s L$ with $4\beta L s$ (nonlinear monomials are atoms).
- A single-step `calc` as the last tactic of a `have` block swallowed the following `exact` line as a further step; use `exact` directly for one-step bounds.

3 Outcome

The $d = 2$ log polynomial is available for the paper's chart integral with ξ, η of class C^2 (via Lipschitz and mixed-difference bounds). Next: the face-parameter second-order theorem under the strict gap, the $C \neq 0$ variants of the leading theorem, and a third Astra consult on all-order subtraction.